

ATHL Fan Token Platform: Functional Requirements - Summary

Overview: ATHL enables self issuance of follower tokens akin to meme coins that go through two phases: (1) fixed step-up pricing until a specific dollar hurdle is reached, followed by (2) open market trading (fans/followers can purchase and trade tokens issued by athletes and creators).

The tokens have transparent purchase patterns, exciting leaderboards, publicly available data and analytics for fantasy competition (amongst our athletes creating their own teams, and between buyers/fans around performance) and robust guardrails and security. For the first 30 days we will only offer the ability to purchase:

1. User Onboarding

1.1 Sign-Up (Issuers and Fans)

- **Purpose:** Register athletes, creators, and fans, verify identities, and set up token issuance/purchasing.
- **Front-End:**
 - Sign-up page (athl.live/signup) with “Athlete,” “Creator,” or “Fan” buttons.
 - Issuer form (multi-step):
 - Personal: Name, email, country, state, zip code, gender, birthdate
 - Athlete: Primary and secondary Sport, position in each sport, level, league, team/school (dropdowns with “Other” text input option).
 - Creator: Type (e.g., musician), content medium (dropdowns with “Other” option).
 - Social media: Primary/secondary platforms (e.g., Instagram, X), handles, up to 5 total social platforms.
 - An open text space to provide more details including: Bio, token perks, video/social media links (clickable),
 - Drop down menu to select the desired raise amount (\$2M–\$10M).
 - For Athletes: “Are you interested in participating in athlete teams to compete in fantasy competitions? Many of these will be run by retired professional athletes.” (Check box)
 - Location: Country, zip code (zip code not shown in profile)
 - Birthday (only age shown in profile)
 - Fan form: Username, email, 2 social handles.
 - Social handles need to be confirmed with OAuth.

- Verification of OnlyFans or other niche social platforms not in OAuth or AuthO.
 - Generate a unique code (e.g., "ATHL-X7B9Q2") and instruct creators to add it to their bio of the target social media website.
 - Scan bio on the platform to make sure it is there. if yes, it is confirmed.
 - Subscribe to onlyfansapi.com (\$99/month), connect via /authenticate, and use /users/{username} to fetch bio data. Poll every 5 seconds to detect the code.
- Username validation:
 - The confirmed social handle with the most followers will be the suggested ATHL user name.
 - Conduct real-time availability check of the username.
 - If their first social handle is already taken
 - Compare social followings of the two handles. If the one trying to create an account has 50X+ the follower size as the other, we may consider letting the larger account use that handle.
 - Otherwise, choose to use the second social handle.
 - If the second social handle is not available, they can create their own. 4-20 alpha numeric.
- **Back-End:**
 - Social verification: OAuth 2.0 (alchemy/Auth0 for Instagram, X, etc.) or challenge code in bio (polled every 5s for 1h). OnlyFans via onlyfansapi.com (\$99/month) or manual screenshot.
 - Identity verification (Socure) for high-profile issuers (>300K followers or pro athletes), fallback challenge code.
 - Email verification (ZeroBounce) to block disposable emails.
 - Create Alchemy wallet, link to account, email funding instructions (USDC, ETH, SOL).
 - Add new issuers to waiting list for launching tokens as they will launch in classes together, grant everyone fan access immediately as long as the platform is launched.
- **Dependencies:** Socure API, OAuth 2.0, onlyfansapi.com, Alchemy, AWS SES.

1.2 Waiting List Management

- **Purpose:** Control issuer token launch onboarding into launch classes/groups. Acceptance to classes will be determined by ATHL Admins. Issuers can immediately act as a fan and purchase other tokens via their fan dashboard.
- **Front-End:**

- Admin page to view/filter/sort issuers by sign-up date, follower count, referrals, purchases.
 - Display of waiting list in a table: User ID, social handle, demographics, issuer type, sport/creator type, token wallet holdings in USD, referrals, referral rewards generated, status, priority designation, notes. This table is sortable and filterable.
 - **Back-End:**
 - Auto-add issuers to waiting list, email confirmation to issuers.
 - Support phased onboarding (100 issuers in Phase 1, 500 in Phase 2, etc.).
 - Automated emails to issuers once they have been accepted to a launch class
 - Manual approval/denial, log actions, export CSV.
 - **Dependencies:** Database, RBAC.
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2. Authentication and Security

2.1 Sign-In

- **Purpose:** Secure user access.
- **Front-End:**
 - Username input (4–20 characters, alphanumeric).
 - Social login (largest or 2nd largest follower platform).
 - Fans: Social (X, Google, Facebook) and email/password, or WebAuthn passkey.
 - MFA prompts: QR code for authenticator apps, biometric, passkey, downloadable backup codes.
 - Error messages for invalid credentials/lockouts.
- **Back-End:**
 - Validate credentials, require MFA for issuers (authenticator app, biometrics, passkey), encourage for fans.
 - Lock accounts after 5 failed attempts (10-min lockout, escalate to 30 min).
 - Log login attempts (timestamp, IP, device), notify lockouts.
 - Session tokens: 30-min inactivity or 24h activity expiry, HTTPS.
- **Dependencies:** OAuth 2.0, WebAuthn, AWS SES, RBAC.

2.2 Security Features

- **Purpose:** Protect platform and user data.
- **Front-End:**
 - CAPTCHA (reCAPTCHA v3) for high-risk purchases (>10/min).
 - Risk score feedback (e.g., “Low risk”), MFA-secured issuer publicity toggle.

- Green check mark on issuer purchaser page for those who completed ID verification (Socure or other)
 - **Back-End:**
 - **Anomaly Detection:** Flag rapid buys (>10/min), same-IP purchases, bot patterns (<100ms clicks) using ML (scikit-learn).
 - **Behavior Biometrics for Bot Detection.**
 - **Delays:** 50–100ms randomized purchase delays during launches.
 - **Risk Scoring of users:** 0–100 score (40% transactional, 30% behavioral, 30% contextual), KYC (Persona) for >70.
 - **Social Monitoring:** Grok API for X, scrape Telegram/Discord/Reddit, correlate with purchase spikes.
 - **Trading Monitoring:** Detect coordinated buys/sells (>3 accounts, >0.05% supply in 10 min), spoofing (>10 canceled orders/min), alert staff.
 - **Reversals:** The ability to reverse trades deemed fraudulent or manipulative. Reversed trades would be seen in user accounts and on ATHLScan (e.g., “Reversed: Bot activity”), We would notify users.
 - **Privacy:** Encrypt data, anonymize PII (GDPR/CCPA), HMAC-SHA256 for trade log integrity.
 - **ATHLScan Protection:** Randomize IDs/codes (each user gets 10-30 anonymous userIDs so no portfolios can be closely tracked. Essentially each portfolio and transaction history is broken up to 10-30 small portfolios/histories), 15 - min timestamp rounding, rate-limit queries (<60/h per IP).
 - **Follower Authenticity:** HypeAuditor for issuers (>10K followers), badge on purchase page (<10% bots).
 - **Dependencies:** Grok, HypeAuditor, Persona, AWS (RDS, Lambda, WAF).
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3. Token Trading Core

3.1 Trading Mechanics

- **Purpose:** Enable token purchases, sales, and transfers. (Purchases only for the first 30 days)
- **Front-End:**
 - Trading interface on issuer purchase page: Input tokens or USD, show estimated cost, select alchemy wallet (USDC, ETH, SOL) or Stripe (CC), confirm with legal agreements.
 - Display limits: ~10,000 tokens/user. We will likely have issuers set their own purchase limits for their fans)
 - Real-time prices.

- **Back-End:**
 - Linear pricing (e.g. from \$1 going to a range of \$3–\$24) for initial sales based on how much issuer wants to raise (from \$2M to \$10M, ATHL pricing will be 25% more, e.g. total raise for ATHL would then be \$2.5M to \$12.5M)
 - Prioritize small purchases (<100 tokens) if large and small orders come in at the same time.
 - Enforce limits via wallet tracking, limit bulk sales, apply 24h price floor (80% prior day's average - adjustable and pause trading if 15% below price floor)
 - Issuer-only transfers to others (up to 10 tokens (admin adjustable), MFA-confirmed), logged as “issuer transfer.”
 - Process trades in <1s, support 10,000 trades/min.
 - Implement dynamic trading pauses when token price dips 15% within 24 hours.

3.2 Order Book and Pricing (Very similar to PolyMarket)

- **Purpose:** Manage secondary market trading and pricing.
 - **Front-End:**
 - Order book summary on purchase page (top 5 buy and sell orders visible but can scroll through entire book).
 - **Back-End:**
 - Maintain real-time order book for secondary trades, support market/limit orders.
 - Calculate prices via supply/demand (midpoint of buy-sell spread).
 - Update prices in real time, log to ATHLScan.
 - **Dependencies:** Database, real-time pricing engine.
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4. Fan Dashboard

- **Purpose:** Central hub for fans to manage tokens and explore issuers. “Fan” should be perceived as a role since all Issuers will be “fans” as well when they purchase or explore other issuer tokens. Issuers will have a fan dashboard in addition to their Issuer Admin page/dashboard.
- **Front-End:**
 - Sections: Wallet balances, balance growth chart, token holdings, 2-3 leaderboards, suggested issuers, Issuer search, watch list, 2-3 team competitions, referrals.
 - Balances: alchemy (non-ATHL) and ATHL (tokens) in USD.
 - Chart: Total balance growth (24h, 1w, 1m, all time), Robinhood style.
 - Holdings: Issuer, price, daily % change, quantity, value; paginated (>10 tokens).

- Suggested issuers to buy: 10 based on sales/growth, fan interests
 - Referral Dashboard showing a graph of income generated so far from referrals, and a table of referees with a breakdown of UserIDs, date of sign up, tokens sold, referral rewards generated, and referral rewards paid so far. Referrals are paid to referring issuers to the platform. Users are paid 2% of tokens sold from referral up to \$1,000. Some people will be under 'agent' contracts and their pay out on initial token sales will be 2% for the first year, 1% for the second year and 0.5% thereafter.)
 - Watch list: Up to 50 issuers, price, % change, remove option.
 - A toggle to choose if they want their portfolio of token holdings publicly viewable via leaderboard competitions. Their user ID will be shown along with portfolio details.
 - Create a Portfolio Results embed engine similar to the embeds for PolyMarket: <https://docs.polymarket.com/polymarket-learn/FAQ/embeds#how-to-use-embeds>, have embed show 1) "ATHL Follower Token Exchange", 2) user ID, 3) current portfolio market cap, 4) portfolio value change over past 24 hours and 7 days. Use orange and white writing on a black background. This embed will primarily go on X and Substack. Users can then copy and paste this into X and Substack posts for live updates on their performance.
 - Real-time updates of all pricing (<15 sec) in their wallet, WCAG 2.1 compliant.
 - Subscription for email notifications: All users can sign-up for a monthly subscription service that provides users real-time alerts to their phones of new issues, and also our analysis of those issuers' socials (e.g. their token at \$XX represents % of their social media): \$9.99/month
 - Transaction history: A table showing all previous transactions the fan made, 20 per page
 - **Back-End:**
 - Fetch balances, prices, issuer data; calculate rankings/suggestions.
 - Log searches, link to purchases for analytics.
 - Persist watch list.
 - **Dependencies:** Chart.js, database
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5. Issuer Pages

5.1 Issuer Token Purchase Page

- **Purpose:** Enable token purchases and profile viewing.
- **Front-End:**
 - Accessible via dashboard or URL (`athl.live/issuer/<issuer_id>`).

- Sections: Price chart (1d, 1w, 1m, all time), sales (\$USD raised, # tokens sold), profile (name, type, verified socials, bio, perks, linked videos, green check if ID verified by Socure), Fan's wallet balance, token purchase window, transaction history of fan for this specific token.
- Purchase window: Tokens/USD input, alchemy/Stripe selection, legal confirmation.
- Issuer's Token Portfolio: A listing of which tokens they are holding, quantity and length of time holding. (some issuers will opt out of this)
- Transaction History: Date, price, amount; paginated (>10).
- Add-to-watch-list, back-to-fan-dashboard button.
- Create an Issuer Token Performance embed engine similar to the embeds for PolyMarket:
<https://docs.polymarket.com/polymarket-learn/FAQ/embeds#how-to-use-embeds>, have embed show 1) "ATHL Follower Token Exchange", 2) user ID and Issuer Name 3) Current Token market cap, 4) Current Token Price, with price change over past 24 hours and 7 days. Use orange and white writing on a black background. This embed will primarily go on X and Substack. Users can then copy and paste this into X and Substack posts for live updates on their performance. Both fans and issuers can post this into X and substack posts.
- **Back-End:**
 - Fetch issuer data, prices, wallet balance.
 - Process purchases, update wallet/history in real-time, log to ATHLScan.
- **Dependencies:** Stripe, database.

5.2 Issuer Admin Pages

- **Purpose:** Monitor sales, edit profiles, manage teams.
- **Front-End:**
 - Header: Name, ATHL UserID, classification (e.g., Athlete, Football).
 - Revenue chart: Daily/cumulative sales (1d, 1w, 6m, 1y, all time) from initial token sales and secondary sales.
 - Metrics: Price, coins sold, coins available, 24h turnover (once trading begins).
 - Breakdown charts: Initial sales only, revenue from fees only.
 - Past and future issuer proceeds table: Previous months + projections (12 months). Each row is a year, columns are months.
 - Editable profile: Sports, schools, team, mission, perks, links (socials read-only).
 - Token holder table: User ID, email, quantity, purchase date, benefits provided (text editable by issuer), benefits due(text editable by issuer); sort/filter.

- Provide issuers the ability to send other users up to 10 of their tokens. Issuers must buy their own tokens first, then send to other users. Users are not allowed to hold on to more than 0.5% of their own tokens.
 - A toggle to switch on open trading of their token. Open trading will occur when a token gets sold out, but can also be switched on earlier to allow early buyers to sell their token at any time.
 - Provide Create an embed engine similar to PolyMarket:
<https://docs.polymarket.com/polymarket-learn/FAQ/embeds#how-to-use-embeds>
 have embed show 1) "ATHL Follower Token Exchange", 2) issuer name, 3) current token price, 4) token price change over 24 hours, 5) Issuer Token Market Cap. Use orange and white writing on a black background. This embed will primarily go on X, Substack and potentially issuer personal websites.
 - Team tables: User ID, email, country, time zone, type, role (Captain, teammate) status, notes, remove (organizer is only one who can edit this).
 - Develop New Team form: Captian IDs (must be issuers 18+yrrs old, up to 5), names, Teammate user IDs, Teammate names, tedam purpose (1000 chars), meeting frequencies, competition frequency, response deadline, contact. Token launch form: Team name, raise (\$2M–\$10M).
 - Internal email system to notify all those who have been invited to the team.
 - Competition request: Team captain challenges other team captains prominent issuers to a challenge. Issuer IDs the captain wants to challenge, details on challenge, date of challenge
 - Internal email sends this out.
 - Challenge is also automatically posted to the ATHL X Account fore everyone to see.
 - **Back-End:**
 - Calculate revenue, projections; save profile edits.
 - Validate team IDs, send email invitations, track responses.
 - Enforce staking (1,000 tokens/leader, 100/member), mint team tokens.
 - Email forwarding for team invites
 - **Dependencies:** Email, escrow, token minting.
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6. Buy Tokens Page

6.1 Buy Tokens

Purpose: Page displays all available tokens to be purchased. Is broken out into athlete and creator and then their sub categories.

- **Front-End:**
 - Shows a grid of issuers showing their name, user ID, photo, athlete/creator and sport/creator details, current price, price change in 24 hours
 - First couple rows are suggested issuers
 - remaining rows are broken into athletes then categories. Within these break out sports, then creator types

6.2 Search / Filter tokens

Purpose: Help fans identify the issuers they are most interested in

- **Front-End:**
 - Provide Universal Search to search through all profiles based on key words
 - Provide filters for athlete / creator and their sub categories.

7. Leaderboard Page

Purpose: Help fans identify trending issuers and macro trends

- **Front-End:**
 - Have 20-30 dashboards overall. Initial ones cover price increase over the last 24 hours and total market cap for all issuers combined, Other dashboards cover similar metrics but broken down by sport and creator categories, men/women and by key country/region (USA, EU) and teams. Updated every minute.

8. Internal Dashboards

8.1 Control Room & Notifications

- **Purpose:** Track signups, referrals, sales and user behavior. Act as the internal control room of the business, connecting directly to backend data sources to enable real-time oversight of platform activity, issuer moderation, risk signals, system health, and compliance workflows.
- **Front-End:**
 - Monitor conversions to identify issues with website, e.g. new visitor converting to sign up converting to actual purchases and referrals
 - Monitor authentication pipeline functionality, accurate & timeline price updates, etc.
 - Compliance alerts: Suspicious patterns, action options.
 - Milestone alerts: In-app/email (e.g., 1,000 tokens sold).
 - Audit log: Admin ID, action, timestamp; filter by admin/date.
- **Back-End:**

- Trigger configurable milestones, detect suspicious trades.
- GDPR/CCPA compliance.
- **Dependencies:** Notification system, Chainalysis.

8.2 KPI and Notifications Dashboard

- **Purpose:** function as business intelligence tools for internal and external teams, offering structured insights into growth, user behavior, token economics, platform usage, and campaign effectiveness—helping stakeholders make informed decisions.
 - **Front-End:**
 - KPIs: Sales (USD/tokens), DAU/WAU, registrations (fans, athletes, creators), referral metrics, revenue (daily, weekly, monthly).
 - Heatmap/table: Purchase/issuance locations, filter by time/issuer/type.
 - Transaction log: ID, date, issuer, fan ID (anonymized), USD, tokens; sort/filter.
 - Repeat purchases: Fan ID, count, USD total, USD avg, date, issuers.
 - Search analysis: Queries, frequency, conversion rate.
 - Segmentation of fans and issuers: High spenders, casual buyers, non-buyers; marketing suggestions.
 - Reports: Custom metrics, timeframes, filters; download CSV/Excel.
 - Issuer Waitlist table: We review issuers and select which class they will launch in via a drop down menu
 - **Back-End:**
 - Fetch real-time data, calculate trends, anonymize PII.
 - Log admin actions, retain 12 months.
 - **Dependencies:** MaxMind, database, reporting engine.
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11 Support / User Feedback

- **Purpose:** Handle user issues and get feedback on features.
- **Front-End:**
 - “Support” link on all pages, form with category, subject, description, transaction ID, attachments, email, priority.
 - “My Tickets” section for status/history.
- **Back-End:**
 - Validate inputs, encrypt data, apply CAPTCHA.
 - Generate ticket ID, email confirmation, allow staff follow-ups.
 - Retain data 12 months.
- **Dependencies:** reCAPTCHA, RBAC.

10. ATHLScan

- **Purpose:** Public transaction transparency.
- **Front-End:**
 - Tab (`athl.live/athlscan`): Table with ID, timestamp, From/To ID (anonymized or public), token, amount, type, payment, fee.
 - Search/filter by ID, token, code, date, type; paginated.
 - FAQ section.
 - ATHLScan Protection: Randomize IDs/codes (each user gets 10-30 anonymous userIDs so no portfolios can be closely tracked. Essentially each portfolio and transaction history is broken up to 10-30 small portfolios/histories), 15 - min timestamp rounding, rate-limit queries (<60/h per IP).
- **Back-End:**
 - Pull from transaction database, exclude PII.
 - Support issuer publicity toggle, 5–30 anonymized codes/user.
 - Log reversals, apply 1% fee, credit issuer wallet.
- **Dependencies:** HMAC-SHA256, AWS WAF.

Non-Functional Requirements

- **Performance:** 10,000 trades/min, dashboard load <2s, support submission <3s.
 - **Backup:** 5-min incremental, daily full, 48h–30d retention, encrypted.
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11 Version 2/3 Features / Functionality

11.1 Fantasy Team Competitions Page

- **Purpose:** Allow fans to view and bet on issuer team events.B
- **Front-End:**
 - List all upcoming competitions: Team names, date, type of competition (e.g., 100m sprint), betting probabilities, “Place Bet” button. Betting very similar to Polymarket format.
 - Betting modal: Team, amount (ATHL tokens), order book, market/limit options.
 - Leaderboard: Top 10 teams by wins, win rate, competitions.
- **Back-End:**
 - Store competition details, maintain order book, update probabilities.
 - Validate balance, lock bets in escrow, payout winners (0.5% ATHL fee).
- **Dependencies:** Escrow, order book system.

11.2 Liquidity Incentives

- **Purpose:** To foster an active and liquid trading environment for most if not all issuers and fans. We want everyone to believe they can get in and out of token positions at any time with relative ease.
 - The system shall reward traders in ATHL who add liquidity to token markets. We will follow a similar reward system to Polymarket which was based on <https://www.dydx.foundation/blog/liquidity-provider-rewards>

ATHL Data Infrastructure Blueprint (includes Dashboards)

https://docs.google.com/document/d/1d-tEFkukj37ly3f8dtX_EUQ5_6hw_vYgeSu2ScXhZa8/edit?usp=sharing